

COMPSCI 589 Final Project - Spring 2026

Due **May 8**, 11:55 pm Eastern Time

Introduction

- Each group member was responsible for evaluating the performance of at least one unique algorithm implemented during the semester on each dataset
- The algorithms tested in this project were chosen from the pool of: k -NN, Decision Trees, multinomial Naive Bayes, Random Forests, and Neural Networks
- All work was done in the shared github repository https://github.com/1985lxw/589_final_project

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- **Katherine** implemented TODO
 - **Tiffany** implemented KNN and Random Forest. She evaluated the hyperparameters k for kNN and n_{tree} for random forest.
 - **Xing-Wei** implemented Decision Trees

Experiments and Analyses

1 The Hand-Written Digits Recognition Dataset

1.1 Neural Networks Algorithm

TODO

1.2 KNN

k	Accuracy	F1 Score
1	0.988571	0.989391
3	0.986857	0.987597
5	0.986286	0.987128
7	0.986286	0.987206
9	0.982857	0.984203
11	0.980571	0.982039
13	0.980000	0.980917
15	0.980571	0.982108
17	0.977143	0.978787
19	0.976000	0.977379
21	0.974857	0.976219
23	0.972571	0.974215
25	0.969714	0.971757
27	0.966286	0.968297
29	0.968571	0.970363
31	0.967429	0.968898
33	0.961714	0.963689
35	0.964571	0.966932
37	0.963429	0.965532
39	0.963429	0.965328
41	0.960571	0.962805
43	0.961143	0.962758
45	0.959429	0.961605
47	0.956571	0.959014
49	0.957714	0.960383
51	0.955429	0.958529



Figure 1: Handwritten Digits Accuracy Distribution

1.3 Random Forest

ntree	1	5	10	20	30	40	50
Accuracy	0.968223	0.969386	0.993427	1.0	0.996774	0.997059	1.0
Precision	0.954737	1.000000	1.000000	1.0	1.000000	1.000000	1.0
Recall	0.980000	0.934332	0.985000	1.0	0.993333	0.994118	1.0
F1 Score	0.962738	0.965294	0.992204	1.0	0.996552	0.996970	1.0

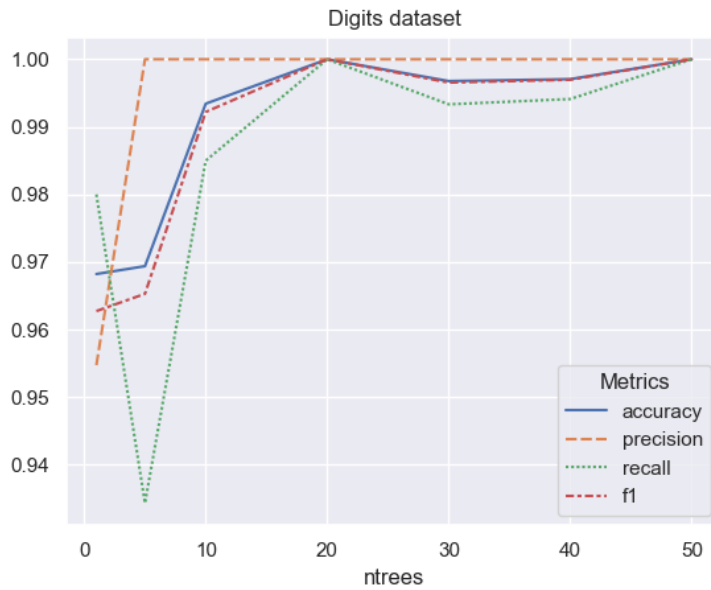


Figure 2: Digits Accuracy and F1 Score Distribution

2 The Oxford Parkinson's Disease Detection Dataset

2.1 Neural Networks Algorithm

TODO

2.2 KNN

k	Accuracy	F1 Score
1	0.966667	0.978033
3	0.944444	0.963738
5	0.944444	0.964514
7	0.944444	0.965475
9	0.927778	0.954590
11	0.900000	0.937926
13	0.872222	0.922714
15	0.861111	0.915862
17	0.850000	0.910674
19	0.838889	0.903946
21	0.811111	0.890880
23	0.827778	0.899946
25	0.850000	0.912620
27	0.838889	0.906579
29	0.838889	0.906613
31	0.844444	0.909247
33	0.866667	0.922270
35	0.872222	0.924905
37	0.872222	0.924716
39	0.850000	0.912842
41	0.855556	0.915269
43	0.833333	0.903978
45	0.844444	0.909831
47	0.822222	0.897769
49	0.827778	0.900780
51	0.816667	0.895134



Figure 3: Handwritten Digits Accuracy Distribution

2.3 Random Forest

ntree	1	5	10	20	30	40	50
Accuracy	0.805556	0.794444	0.816667	0.833333	0.850000	0.838889	0.850000
Precision	0.836293	0.843709	0.853481	0.853754	0.861730	0.854793	0.872728
Recall	0.935714	0.907143	0.928571	0.950000	0.964286	0.957143	0.950000
F1 Score	0.881610	0.872319	0.887543	0.898543	0.909364	0.902490	0.908122

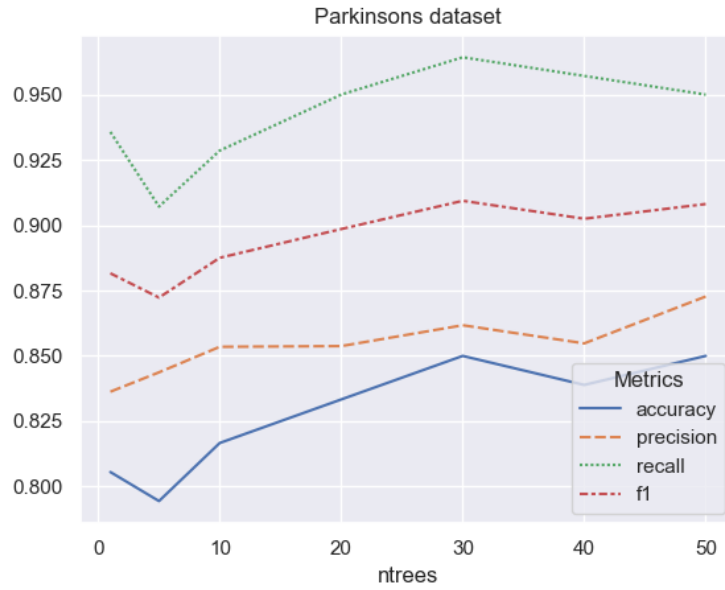


Figure 4: Parkinsons Accuracy and F1 Score Distribution

3 The Rice Grain Dataset

3.1 Neural Networks Algorithm

TODO

3.2 KNN

k	Accuracy	F1 Score
1	0.886614	0.867819
3	0.909974	0.894640
5	0.918898	0.904873
7	0.923885	0.910538
9	0.922047	0.908377
11	0.922835	0.909521
13	0.924409	0.911221
15	0.925984	0.913105
17	0.925984	0.912772
19	0.927034	0.914136
21	0.926509	0.913226
23	0.927034	0.914092
25	0.928084	0.915473
27	0.925984	0.912969
29	0.925459	0.912357
31	0.928346	0.915703
33	0.926509	0.913735
35	0.925984	0.912977
37	0.926509	0.913497
39	0.927559	0.915076
41	0.927034	0.914137
43	0.927559	0.914960
45	0.924934	0.911821
47	0.927559	0.914945
49	0.927559	0.914773
51	0.928084	0.915623



Figure 5: Rice Accuracy and F1 Score Distribution

3.3 Decision Trees Algorithm

We chose to test decision trees on the rice dataset because it has a moderate number of features (7) and a relatively large number of samples (3810), which allows us to explore how different tree depths and minimum split sizes affect performance.

Tuning for dataset with 3810 rows and 7 features.

Testing Depths: [2, 3, 5, 7, 9, 11]

Testing Splits: [190, 381, 571, 762]

Depth	Split	Test Acc	Test F1
2	190	0.9215	0.9192
2	381	0.9228	0.9205
2	571	0.9226	0.9202
2	762	0.9239	0.9215
3	190	0.9234	0.9210
3	381	0.9213	0.9189
3	571	0.9202	0.9179
3	762	0.9239	0.9215
5	190	0.9213	0.9189
5	381	0.9234	0.9210
5	571	0.9226	0.9202
5	762	0.9239	0.9215
7	190	0.9223	0.9199
7	381	0.9226	0.9203
7	571	0.9215	0.9192
7	762	0.9239	0.9215
9	190	0.9234	0.9210
9	381	0.9220	0.9197
9	571	0.9239	0.9215
9	762	0.9239	0.9215
11	190	0.9223	0.9199
11	381	0.9220	0.9197
11	571	0.9218	0.9194
11	762	0.9239	0.9215

=====
BEST CONFIGURATION:

Max Depth: 11

Min Split: 762

F1 Score: 0.9215
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Final Results:

Training Accuracy: 0.9239

Training F1 Score: 0.9216

Testing Accuracy: 0.9239

Testing F1 Score: 0.9215

For both models, the training scores are highly concentrated around 0.92, while the testing scores are more spread out, with a mean around 0.923 for accuracy and 0.922 for F1 score. The training performance suggests that the models have high stability and low bias, however, the testing performance indicates that there may be some sensitivity to the specific training data used in each fold. Since the mean training scores are very close to the mean testing scores, it suggests that the models are robust and are not overfitting. Decision trees can easily

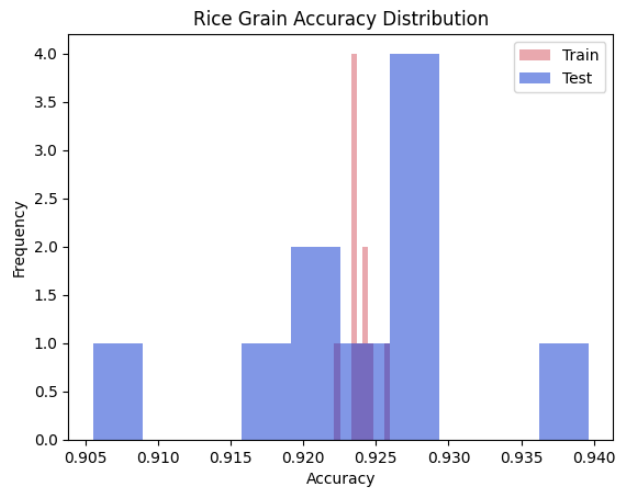


Figure 6: Rice Accuracy Distribution

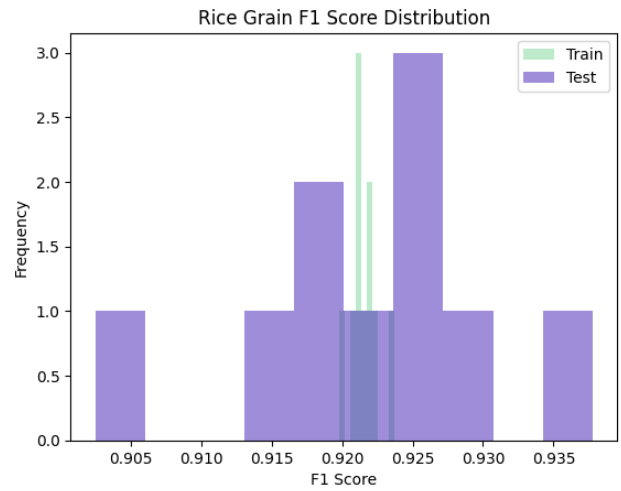


Figure 7: Rice F1 Score Distribution

split numerical features to find precise boundaries, so they are well-suited for the rice dataset, which consists of numerical attributes.

4 The Credit Approval Dataset

4.1 Neural Networks Algorithm

TODO

4.2 KNN

k	Accuracy	F1 Score
1	0.806250	0.780158
3	0.859375	0.845093
5	0.864062	0.853219
7	0.873437	0.864596
9	0.862500	0.854968
11	0.865625	0.858536
13	0.856250	0.848834
15	0.860938	0.854310
17	0.862500	0.858192
19	0.864062	0.860247
21	0.865625	0.860856
23	0.857812	0.853120
25	0.864062	0.858874
27	0.867188	0.863106
29	0.857812	0.854402
31	0.860938	0.856263
33	0.868750	0.865578
35	0.857812	0.852417
37	0.862500	0.857252
39	0.864062	0.856573
41	0.860938	0.853287
43	0.856250	0.848844
45	0.862500	0.855567
47	0.860938	0.854175
49	0.864062	0.856989
51	0.856250	0.849728



Figure 8: Credit Approval Accuracy and F1 Score Distribution

4.3 Decision Trees Algorithm

Tuning for dataset with 653 rows and 15 features.

Testing Depths: [2, 4, 6, 8, 10, 12]

Testing Splits: [32, 65, 97, 130]

Depth	Split	Test Acc	Test F1
2	32	0.8622	0.8621
2	65	0.8639	0.8636
2	97	0.8622	0.8622
2	130	0.8622	0.8620
4	32	0.8378	0.8358
4	65	0.8482	0.8473
4	97	0.8588	0.8580
4	130	0.8577	0.8566
6	32	0.8318	0.8293
6	65	0.8302	0.8288
6	97	0.8501	0.8491
6	130	0.8610	0.8606
8	32	0.8378	0.8356
8	65	0.8576	0.8568
8	97	0.8496	0.8489
8	130	0.8620	0.8612
10	32	0.8377	0.8356
10	65	0.8409	0.8397
10	97	0.8437	0.8423
10	130	0.8504	0.8495
12	32	0.8375	0.8350
12	65	0.8451	0.8439
12	97	0.8668	0.8654
12	130	0.8635	0.8626

=====

BEST CONFIGURATION:

Max Depth: 12

Min Split: 97

F1 Score: 0.8654

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Final Results:

Training Accuracy: 0.8746

Training F1 Score: 0.8741

Testing Accuracy: 0.8606

Testing F1 Score: 0.8601

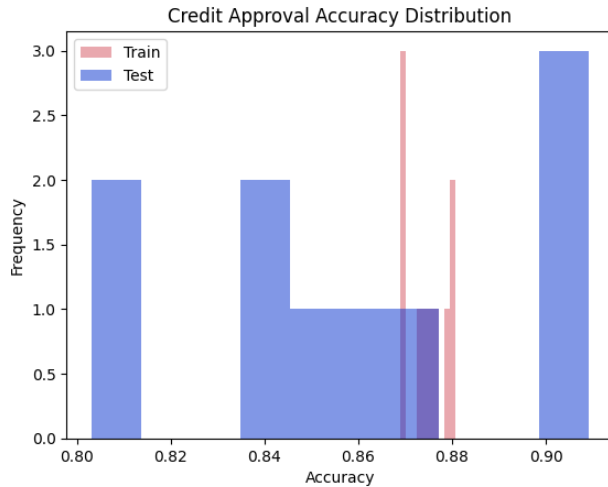


Figure 9: Credit Approval Accuracy Distribution

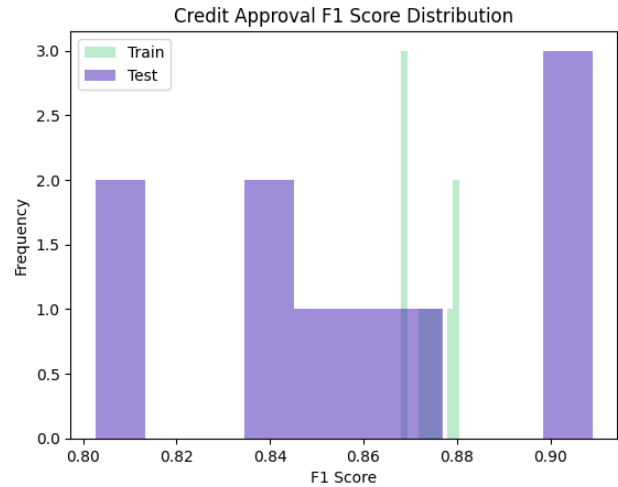


Figure 10: Credit Approval F1 Score Distribution

	Handwritten Digits		Parkinson's		Rice Grains		Credit Approval	
	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score
Neural Networks								
K-Nearest Neighbors								
Decision Trees	0.8344	0.8327	0.8508	0.8101	0.9239	0.9215	0.8606	0.8601

Extra Credit

We did the following extra credit tasks. Describe what you did as part of that task and present the corresponding results. Any source code you develop should be included in your Gradescope submission.

[QE.1] [Extra Credit Question #1] (10 Points)

You may earn up to 10 extra points if you analyze all four datasets using an additional algorithm. In other words, you would be evaluating more than two algorithms on each of the datasets discussed in Section 1.

[QE.2] [Extra Credit Question #2] (10 Points)

You may earn up to 10 extra points if you select a new challenging dataset and evaluate the performance of different algorithms on it. A challenging dataset should necessarily include both categorical and numerical attributes and have more than two classes; or it should be a dataset where you have to classify images.

[QE.3] [Extra Credit Question #3] (15 Points)

You may earn up to 15 extra points if you combine different algorithms and construct an ensemble. For instance, you could create an ensemble composed of three neural networks (each with a different architecture) and a random forest. The ensemble would be trained as described in class: each algorithm would be trained on its own bootstrap dataset, and so on. The final predictions would then be determined by majority voting. Evaluate your ensemble algorithm on all four datasets discussed in Section 1.

[QE.3] [Extra Credit Question #3] (15 Points)

You may earn up to 15 extra points if you implement a new type of algorithm—a non-trivial variant of one of the algorithms studied in class. If you choose to do so, you should evaluate its performance on all four datasets.